

Initial-Value Problems (IVP)

an initial-value problem gives you a problem to solve with a subject restriction

$$\text{Solve } \frac{dy}{dx} = f(x, y) \quad \leftarrow \text{solve this}$$

$$\text{subject to } y(x_0) = y_0 \quad \leftarrow \text{restriction}$$

examples:

$$y = \frac{1}{1 + c_1 x}, \quad y(0) = -\frac{1}{3}$$


• this is a one-parameter family of solution because it only has one c_1 . if it has two c 's (the c 's are called parameters) it would be called a two-parameter family of solutions

$$y = c_1 e^x + c_2 e^{-x}, \quad y(0) = 1, \quad y'(0) = 2$$


this has two c 's in it so we call this a two-parameter family of solutions

Practice Problems

Suppose $y = \frac{1}{1 + c_1 e^x}$ is a one-parameter family of solution of 1st-order
Find the IVP

a) $y(0) = -\frac{1}{3}$



• Plug the x and y into the equation and solve for c_1

$$y = \frac{1}{1+c_1 e^{-x}} \Rightarrow -\frac{1}{3} = \frac{1}{1+c_1 e^0} \Rightarrow -\frac{1}{3} = \frac{1}{1+c_1}$$

$$-\frac{1}{3}(1+c_1) = 1 \Rightarrow -\frac{1}{3} - \frac{1}{3}c_1 = 1 \Rightarrow \frac{-1-c_1}{3} = 1$$

$$c_1 = -4$$

Plug the c_1 back into the "y" equation and that will be your final answer

$$y = \frac{1}{1-4e^{-x}}$$

Solve $y = \frac{1}{x^2+c}$, Find the IVP if $y(-2) = 1/2$

$$y(-2) = 1/2$$

↑ x
↑ y

Plug x and y into the equation and solve for c

$$y = \frac{1}{x^2+c} \Rightarrow \frac{1}{2} = \frac{1}{(-2)^2+c} \Rightarrow \frac{1}{2} = \frac{1}{4+c} \Rightarrow \frac{1}{2}(4+c) = 1$$

$$2 + \frac{c}{2} = 1 \Rightarrow \frac{c}{2} = -1 \Rightarrow c = -2$$

• Plug back into "y" and that will be your final answer

$$y = \frac{1}{x^2-2}$$

Domain:

$$x^2 - 2 \neq 0$$

$$x^2 \neq 2$$

$$x \neq \pm\sqrt{2}$$

$$(-\infty, -\sqrt{2}) \cup (-\sqrt{2}, +\sqrt{2}) \cup (+\sqrt{2}, \infty)$$

General solution is $y = c_1 e^x + c_2 e^{-x}$ Find the IVP for $y(0)=1$, $y'(0)=2$

this problem is a two-parameter solution also you will need to take the derivative for one of them

$$y(0)=1, \quad y'(0)=2$$

$\begin{matrix} \uparrow & \uparrow \\ x & y \end{matrix}$

$$y' = c_1 e^x - c_2 e^{-x}$$

• we take the derivative because one of the inputs requires that we take the derivative

$$\begin{aligned} y(0)=1 & \quad \begin{cases} c_1 e^0 + \cancel{c_2 e^0} = 1 \\ c_1 e^0 - \cancel{c_2 e^0} = 2 \end{cases} + \\ y'(0)=2 & \quad \begin{cases} c_1 e^0 - \cancel{c_2 e^0} = 2 \end{cases} \\ \hline 2c_1 = 3 & \Rightarrow \boxed{c_1 = 3/2} \end{aligned}$$

we have now found c_1 we can plug them into any of the two following equation and solve for c_2

$$c_1 e^0 + c_2 e^0 = 1 \Rightarrow \left(\frac{3}{2}\right) + c_2 = 1 \Rightarrow \boxed{c_2 = -1/2}$$

plug c_1 and c_2 into "y" and that will be your final answer

$$\boxed{y = \frac{3}{2} e^x - \frac{1}{2} e^{-x}}$$

• this is the IVP

General solution is $y = c_1 e^x + c_2 e^{-x}$ Find the IVP for $y(1) = 0$, $y'(1) = e$

$$y(1) = 0, \quad y'(1) = e$$

(Note: In the original image, blue arrows point from 'x' to the '1' in both equations, and red arrows point from 'y' to the '1' in both equations.)

$$y' = c_1 e^x - c_2 e^{-x}$$

• We need to take the derivative because one of the inputs requires us to do so

$$\begin{aligned} y(1) = 0 & \quad \begin{cases} c_1 e^1 + \cancel{c_2 e^1} = 0 \\ c_1 e^1 - \cancel{c_2 e^1} = e \end{cases} \\ y'(1) = e & \quad \begin{cases} c_1 e^1 - \cancel{c_2 e^1} = e \end{cases} \end{aligned}$$
$$\frac{2c_1 e}{2c_1 e} = \frac{e}{e} \Rightarrow \boxed{c_1 = \frac{1}{2}}$$

We already found c_1 now we can plug c_1 into any equation and find c_2

$$c_1 e + c_2 e^{-1} = 0 \Rightarrow \left(\frac{1}{2}\right)e + c_2 e^{-1} = 0 \Rightarrow c_2 e^{-1} = -\frac{e}{2}$$

$$\boxed{c_2 = -\frac{e^2}{2}}$$

Plug c_1 and c_2 into the "y" equation and that will be your final answer

• this is the IVP

$$\boxed{y = \frac{1}{2}e^x - \frac{e^2}{2}e^{-x}}$$

$$y = c_1 \cos(2x) + c_2 \sin(2x)$$

• this is a two-parameter solution because it has two C's

$$y(0) = 1, \quad y'(\pi) = 5$$

↑
x ↑
y

$$y' = -2c_1 \sin(2x) + 2c_2 \cos(2x)$$

• we take the derivative because one of the inputs requires us to do so

$$\begin{cases} y(0) = 1 & c_1 \cos(2 \cdot 0) + c_2 \sin(2 \cdot 0) = 1 \\ y'(\pi) = 5 & -2c_1 \sin(2 \cdot \pi) + 2c_2 \cos(2 \cdot \pi) = 5 \end{cases}$$

$$\begin{cases} y(0) = 1 & \begin{cases} c_1 + 0 = 1 \\ 0 + 2c_2 = 5 \end{cases} \Rightarrow \boxed{c_1 = 1} \\ y'(\pi) = 5 & \end{cases} \Rightarrow \boxed{c_2 = 5/2}$$

plug c_1 and c_2 into the "y" and that will be your final answer

$$\boxed{y = \cos(2x) + 5/2 \sin(2x)}$$

• this is the IVP